

Background

- LLMs excel at math reasoning (DeepSeek-R1, etc.) but are expensive to deploy
- Knowledge Distillation: compress large model (teacher) → small model (student)
- Why math reasoning? Requires multi-step logical deduction, a core LLM capability benchmark

Terminologies

Chain-of-Thought (CoT) : Step-by-step reasoning traces generated by the teacher model.

Temperature : Controls the randomness of token sampling during generation. Low T prioritizes correctness; high T encourages diversity and exploration.

Semantic Deduplication : Removing near-duplicate training samples by computing cosine similarity between answer embeddings via Sentence-BERT.

Log-Probability Confidence : The per-token average log-probability $\bar{\ell} = \frac{1}{L} \sum \log P(y_t | y_{<t}, p)$ of a generated CoT trace, measuring how certain the teacher is about its output.

Distillation Tax : The phenomenon where fine-tuning an instruction-tuned model on foreign-style CoT traces degrades its native reasoning performance, rather than improving it.

Task

- **Goal**: Distill math reasoning from DeepSeek-R1-Distill-Qwen-1.5B → Qwen2.5-0.5B
- **Datasets**: GSM8K (7,473 train / 1,319 test) + MATH (11,248 train / 1,250 test)
- **Evaluation**: Final-answer accuracy

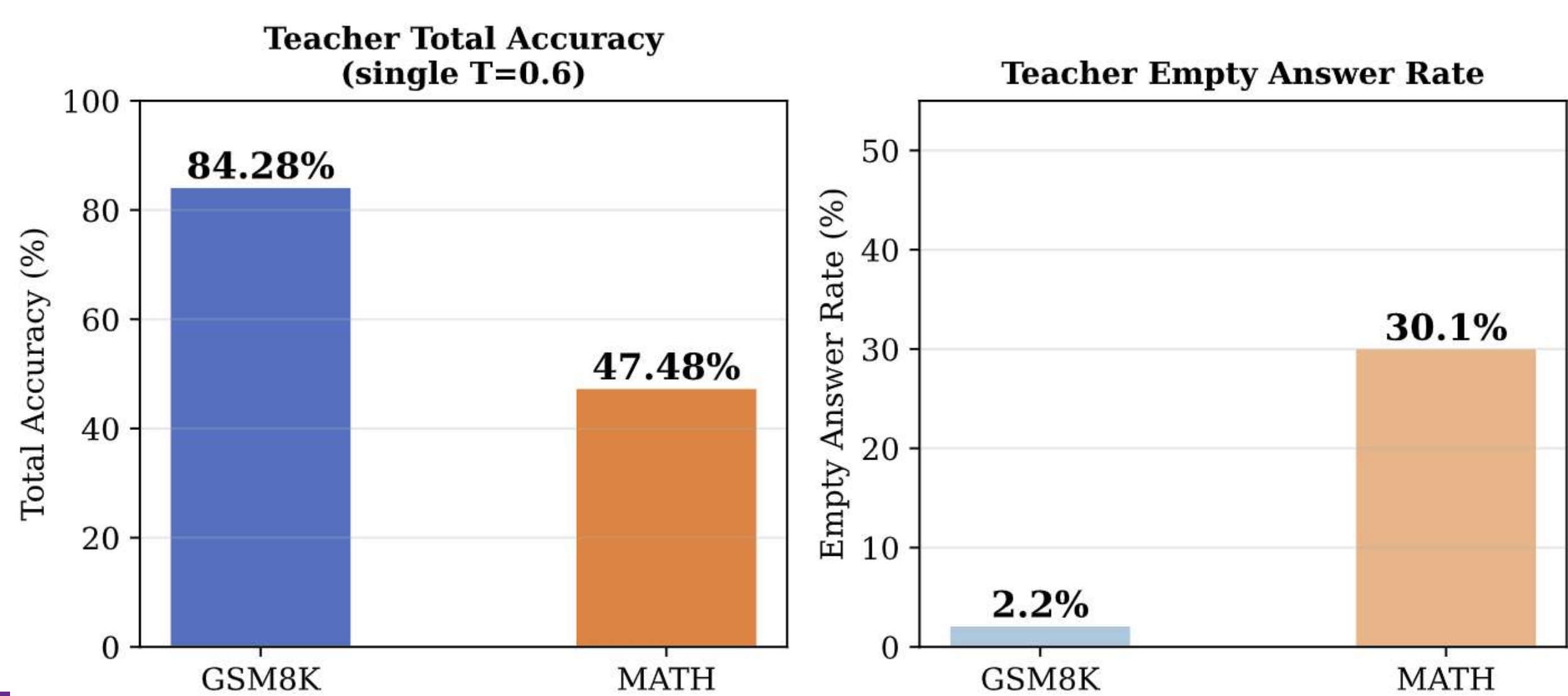
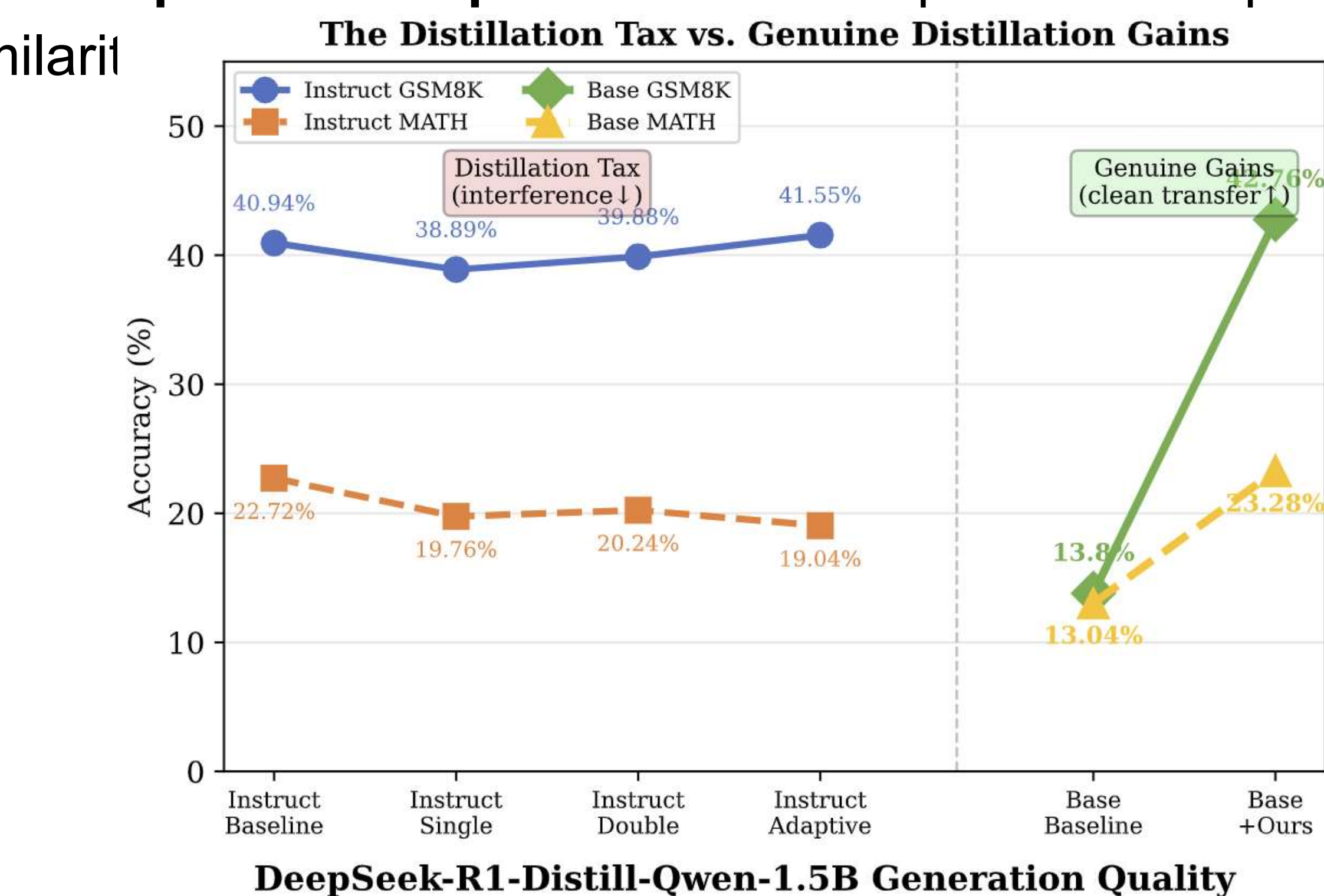
Challenges

Quality vs. Diversity : Low T → lacks diversity; High T → noisy

Distillation Tax : Instruct models underperform after CoT fine-tuning

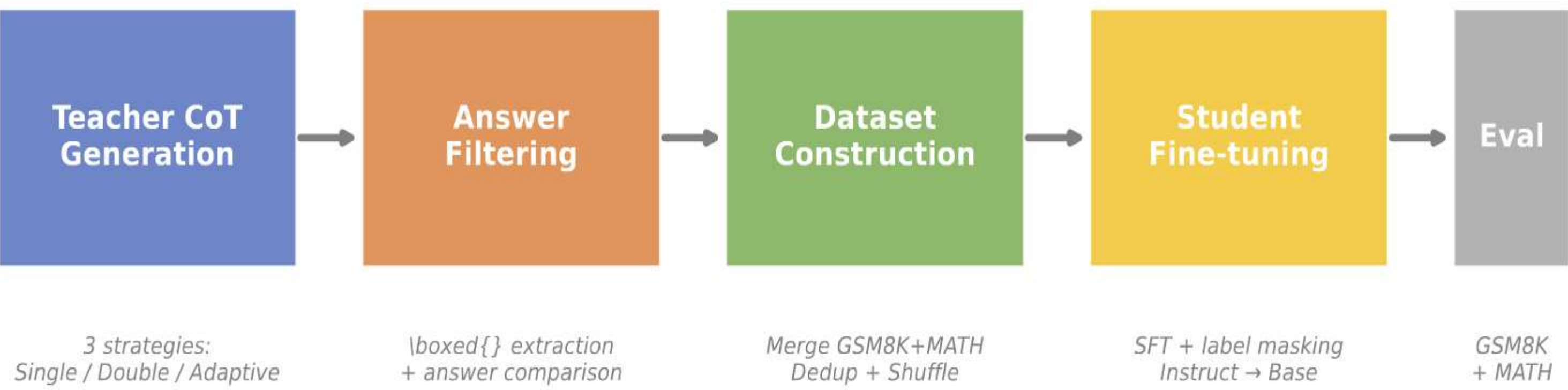
Unreliable Teacher Outputs : MATH 30% empty answers, token truncation. Unformatted answers.

Near-Duplicate Samples : 70% same-problem sample pairs ≥ 0.92 similar!



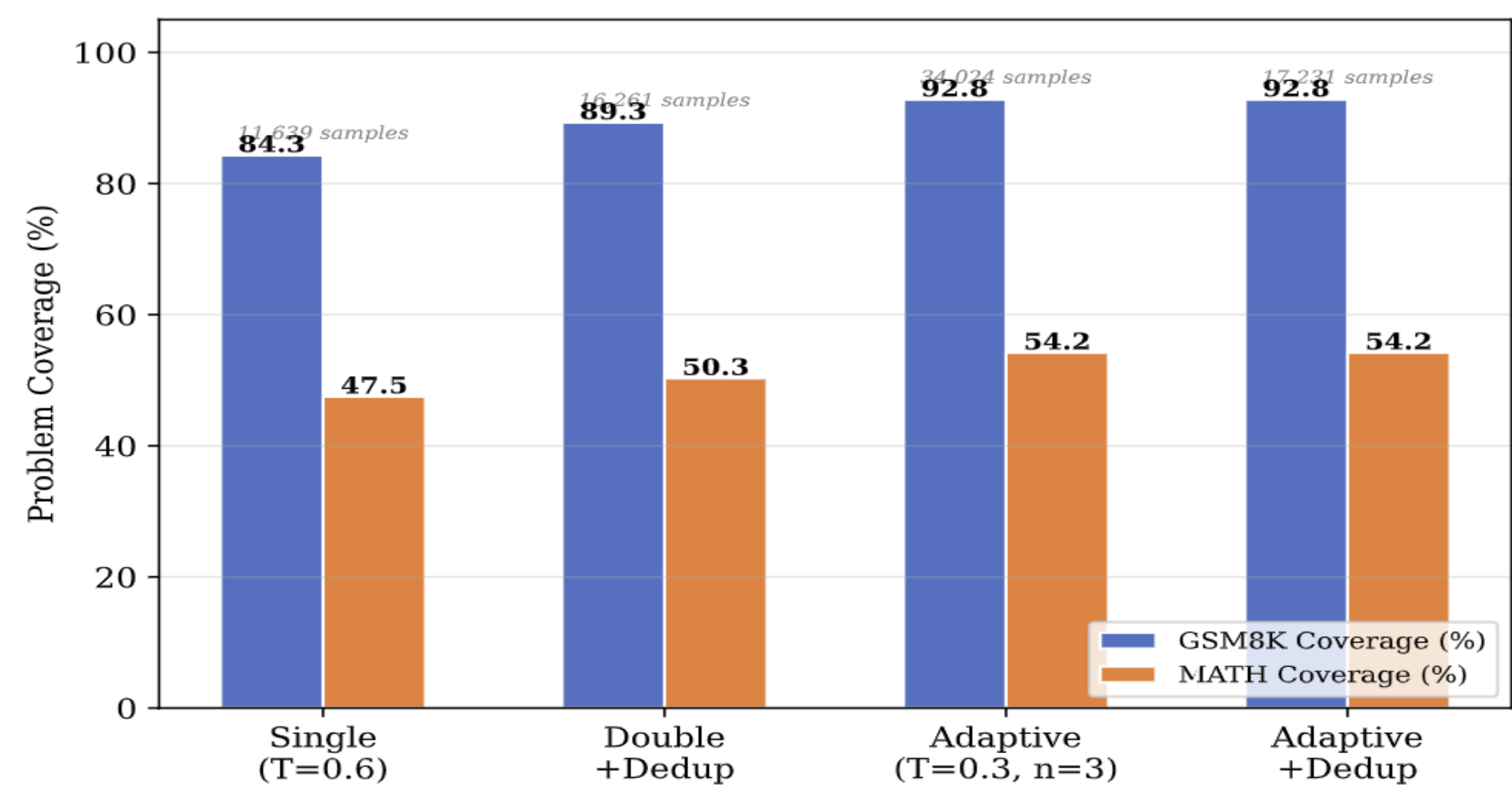
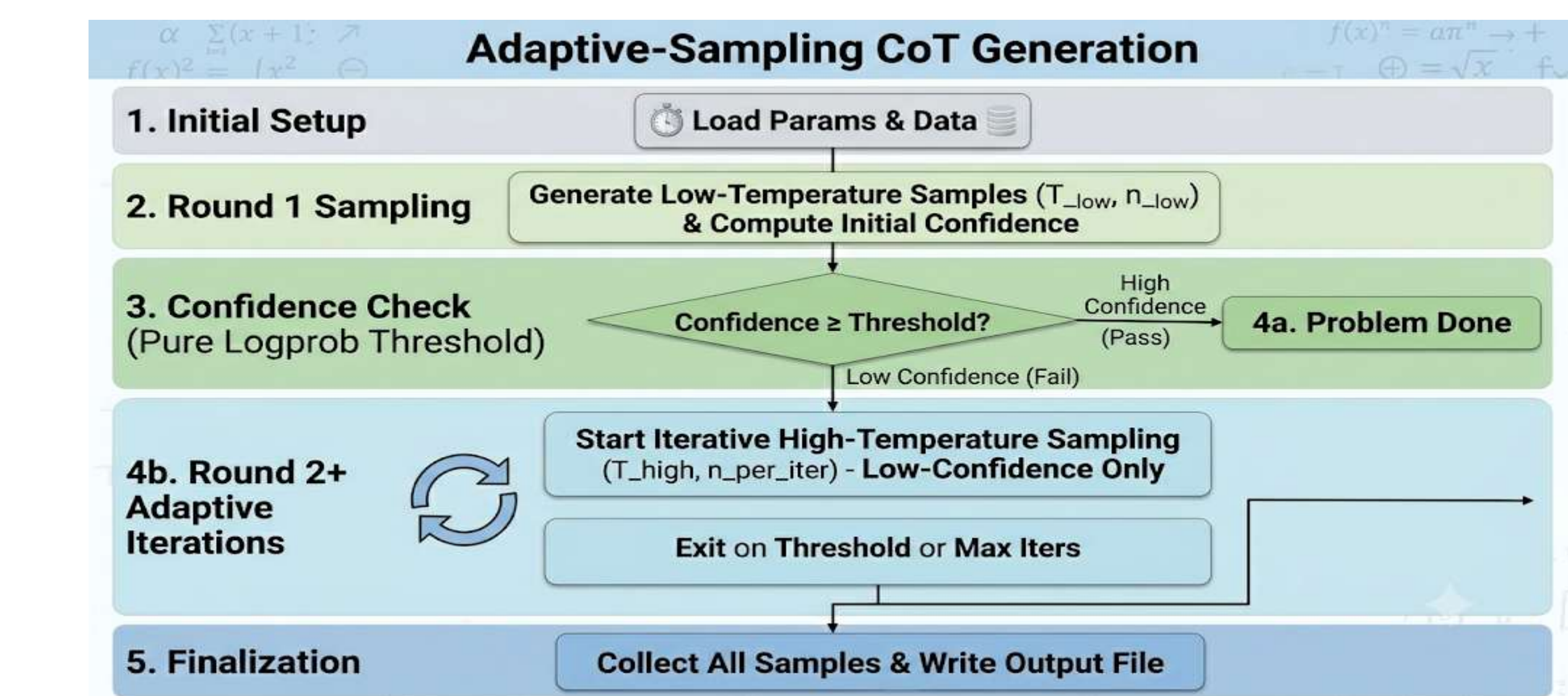
Our Approach

Knowledge Distillation Pipeline



Three Strategies:

1. Single-Temp (T=0.6) — baseline dataset
2. Double-Temp (T=0.6 + 0.9) + Semantic Dedup
3. Adaptive (T=0.3, n=3) (+ Dedup)



Experiments

Setup

Datasets:

GSM8K (7,473 train / 1,319 test) and MATH (11,248 train / 1,250 test)

Metrics: Final-answer accuracy.

Teacher Model: DeepSeek-R1-Distill-Qwen-1.5B.

Student Model: Qwen2.5-0.5B. SFT with label masking

Compared Models

Instruct Line:

- Instruct Baseline — Qwen2.5-0.5B-Instruct, zero-shot
- Instruct-Single — SFT on single-temp
- Instruct-Double — SFT on double-temp + dedup
- Instruct-Adaptive — SFT on adaptive dataset, no dedup

Base Line:

- Base Baseline — Qwen2.5-0.5B, zero-shot
- Base-Adaptive+Dedup — SFT on adaptive + dedup dataset

Results

